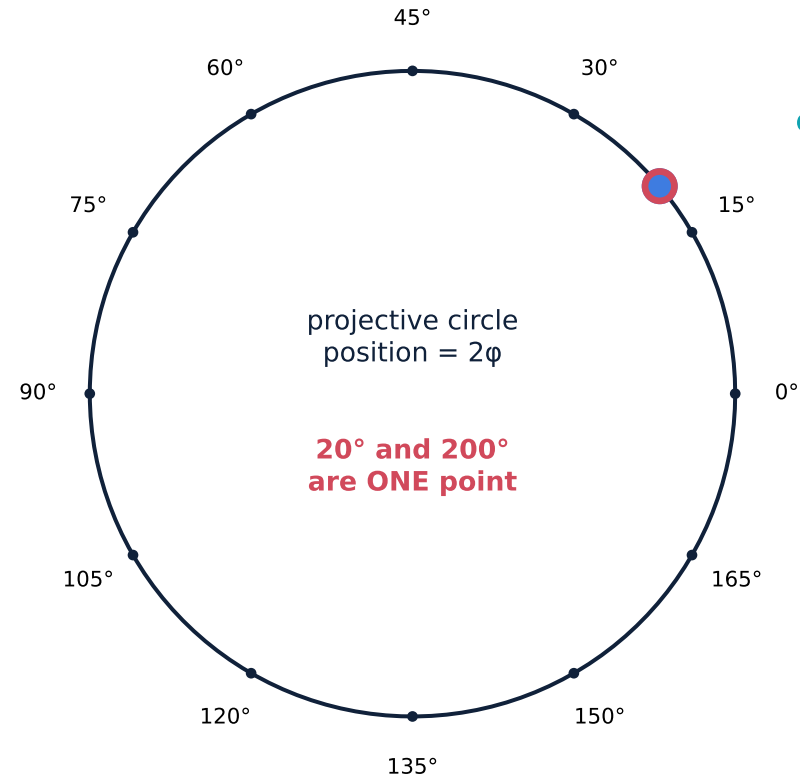
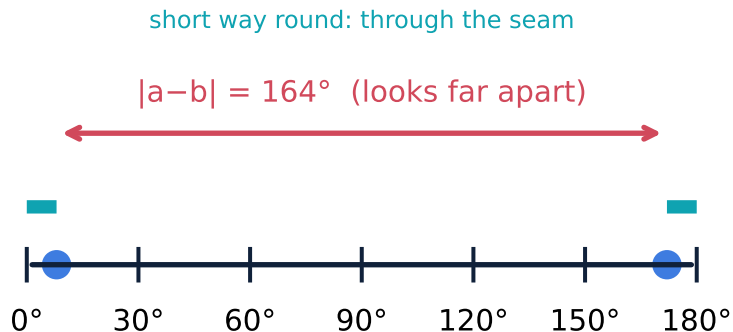


## Unoriented directions live on $[0, \pi)$



## Wrap-around: a plain interval is not enough

$$d_\pi(a,b) = \min(|a-b|, 180^\circ - |a-b|) = 16^\circ \rightarrow \text{actually adjacent!}$$



$0^\circ$  and  $180^\circ$  are glued: the interval wraps around